

ROS2 day_4 hw1 Report

로봇 21기 예비단원 최동현

```
choi@choi-15Z90Q-GRLGL: ~/ros2_ws
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [fake_imu-1]: process started with pid [10050]
[INFO] [imu_watchdog-2]: process started with pid [10051]
[INFO] [ros2-3]: process started with pid [10095]
[INFO] [ros2-4]: process started with pid [10096]
[fake_imu-1] [INFO] [1789716922.198561081] [fake_imu]: configured: rate=50.0Hz, offset=0.00s
[ros2-3] Transitioning successful
[imu_watchdog-2] [INFO] [1789716922.207363651] [imu_watchdog]: configured: timeout=0.50s, check_rate=10.0Hz
[ros2-4] Transitioning successful
[INFO] [ros2-3]: process has finished cleanly [pid 10095]
[INFO] [ros2-4]: process has finished cleanly [pid 10096]
[INFO] [ros2-5]: process started with pid [10125]
[INFO] [ros2-6]: process started with pid [10126]
[fake_imu-1] [INFO] [1789716923.698502987] [fake_imu]: activated: /imu 발행 시작
[ros2-5] Transitioning successful
[imu_watchdog-2] [INFO] [1789716923.701012922] [imu_watchdog]: activated: /imu 감시 시작
[ros2-6] Transitioning successful
[INFO] [ros2-6]: process has finished cleanly [pid 10126]
[INFO] [ros2-5]: process has finished cleanly [pid 10125]
choi@choi-15Z90Q-GRLGL: ~/ros2_ws$ ros2 lifecycle set /fake_imu deactivate # → 위치독: fake_imu state: inactive
Transitioning successful
choi@choi-15Z90Q-GRLGL: ~/ros2_ws$ ros2 lifecycle set /fake_imu deactivate # → 위치독: fake_imu state: inactive
Transitioning successful
choi@choi-15Z90Q-GRLGL: ~/ros2_ws$
```

<실행결과 1>

그룹을 나누지 않았을 때이다.
콜백을 계속 기다리는 상황

```
int main(int argc, char ** argv)
{
    rclcpp::init(argc, argv);
    auto node = std::make_shared<ImuWatchdog>();
    rclcpp::executors::MultiThreadedExecutor exec;
    exec.add_node(node->get_node_base_interface());
    exec.spin();
    rclcpp::shutdown();
    return 0;
}
```

multi Thread로 바꾸니깐 해결되었다.

```

choi@choi-15Z90Q-GRLGL: ~/ros2_ws
[imu_watchdog-2] [INFO] [1789717611.587624182] [imu_watchdog]:
fake_imu state: active
[imu_watchdog-2] [INFO] [1789717612.588126750] [imu_watchdog]:
fake_imu state: active
[imu_watchdog-2] [INFO] [1789717613.587959049] [imu_watchdog]:
fake_imu state: active
[imu_watchdog-2] [INFO] [1789717614.588103592] [imu_watchdog]:
fake_imu state: active
[imu_watchdog-2] [INFO] [1789717615.587993165] [imu_watchdog]:
fake_imu state: active
[imu_watchdog-2] [INFO] [1789717616.587748972] [imu_watchdog]:
fake_imu state: active
[imu_watchdog-2] [INFO] [1789717617.587991940] [imu_watchdog]:
fake_imu state: active
[imu_watchdog-2] [INFO] [1789717618.587742005] [imu_watchdog]:
fake_imu state: active
[fake_imu-1] [INFO] [1789717618.726568548] [fake_imu]: deactivate
ted: /imu 발행 중단
[imu_watchdog-2] [ERROR] [1789717619.286739718] [imu_watchdog]:
[NO DATA] 0.58s 동안 /imu 수신 없음
[imu_watchdog-2] [INFO] [1789717619.587887633] [imu_watchdog]:
fake_imu state: inactive
[imu_watchdog-2] [INFO] [1789717619.587887633] [imu_watchdog]:
fake_imu state: inactive
- cleanup [2]
- activate [3]
- shutdown [6]
choi@choi-15Z90Q-GRLGL:~/ros2_ws$ ros2 lifecycle set /fake_im
u deactivate # → 위치독: fake_imu state: shutdown
Unknown transition requested, available ones are:
- cleanup [2]
- activate [3]
- shutdown [6]
choi@choi-15Z90Q-GRLGL:~/ros2_ws$ ros2 lifecycle set /fake_im
u shutdown # → 위치독: fake_imu state: shutdown
Transitioning successful
choi@choi-15Z90Q-GRLGL:~/ros2_ws$ ros2 lifecycle set /fake_im
u activate
Unknown transition requested, available ones are:
choi@choi-15Z90Q-GRLGL:~/ros2_ws$ ros2 lifecycle set /fake_im
u cleanup
Unknown transition requested, available ones are:
choi@choi-15Z90Q-GRLGL:~/ros2_ws$ ^C
choi@choi-15Z90Q-GRLGL:~/ros2_ws$ ^C
choi@choi-15Z90Q-GRLGL:~/ros2_ws$ ros2 lifecycle set /fake_im
u deactivate # → 위치독: fake_imu state: inactive
Transitioning successful
choi@choi-15Z90Q-GRLGL:~/ros2_ws$

```

<실행 결과 2>

activate상태에서 inactive로 전이한 상황

```

choi@choi-15Z90Q-GRLGL: ~/ros2_ws
[imu_watchdog-2] [INFO] [1789717536.341494180] [imu_watchdog]:
fake_imu state: inactive
[imu_watchdog-2] [INFO] [1789717537.342021182] [imu_watchdog]:
fake_imu state: inactive
[imu_watchdog-2] [INFO] [1789717538.341636267] [imu_watchdog]:
fake_imu state: inactive
[imu_watchdog-2] [INFO] [1789717539.341736118] [imu_watchdog]:
fake_imu state: inactive
[imu_watchdog-2] [INFO] [1789717540.340966224] [imu_watchdog]:
fake_imu state: inactive
[fake_imu-1] [INFO] [1789717540.418748119] [fake_imu]: shutdown
[imu_watchdog-2] [INFO] [1789717541.341828963] [imu_watchdog]:
fake_imu state: finalized
[imu_watchdog-2] [INFO] [1789717542.342326133] [imu_watchdog]:
fake_imu state: finalized
[imu_watchdog-2] [INFO] [1789717543.341963741] [imu_watchdog]:
fake_imu state: finalized
[imu_watchdog-2] [INFO] [1789717544.341538132] [imu_watchdog]:
fake_imu state: finalized
[imu_watchdog-2] [INFO] [1789717545.341970115] [imu_watchdog]:
fake_imu state: finalized
[imu_watchdog-2] [INFO] [1789717546.342151384] [imu_watchdog]:
fake_imu state: finalized
[imu_watchdog-2] [INFO] [1789717546.342151384] [imu_watchdog]:
fake_imu state: finalized
u deactivate # → 위치독: fake_imu state: inactive
Transitioning successful
choi@choi-15Z90Q-GRLGL:~/ros2_ws$ ros2 lifecycle set /fake_im
u deactivate # → 위치독: fake_imu state: clean
Unknown transition requested, available ones are:
- cleanup [2]
- activate [3]
- shutdown [6]
choi@choi-15Z90Q-GRLGL:~/ros2_ws$ ros2 lifecycle set /fake_im
u deactivate # → 위치독: fake_imu state: cleanup
Unknown transition requested, available ones are:
- cleanup [2]
- activate [3]
- shutdown [6]
choi@choi-15Z90Q-GRLGL:~/ros2_ws$ ros2 lifecycle set /fake_im
u deactivate # → 위치독: fake_imu state: shutdown
Unknown transition requested, available ones are:
- cleanup [2]
- activate [3]
- shutdown [6]
choi@choi-15Z90Q-GRLGL:~/ros2_ws$ ros2 lifecycle set /fake_im
u shutdown # → 위치독: fake_imu state: shutdown
Transitioning successful
choi@choi-15Z90Q-GRLGL:~/ros2_ws$

```

<실행결과 3>

inactive상태에서 finalized로 전이한 상황